

Kyle Lee

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Education

Carnegie Mellon University, Pittsburgh, PA

May 2026

B.S. Computer Science | Concentration in Computer Systems

GPA: 3.96/4.00

- **Selected Coursework:** Cloud Computing, Distributed Systems, Database Systems, Parallel Computer Architecture, Generative AI, Introduction to Machine Learning, Computer Vision, Computer Security

Work Experience

Software Engineer Intern | Stripe

May 2025 – Aug 2025

- Automated database scaling to handle first of month subscription renewals, reducing toil by 75%
- Developed primitive in Go enabling team to reliably schedule database vertical scaling operations
- Increased infrastructure reliability by designing and implementing AWS EBS database autoscaling
- Designed tooling to automate AWS EC2 capacity requests, reducing manual operations from 2,000+ to 1

Software Engineer Intern | Penrose Carnegie Mellon

May 2024 – Aug 2024

- Decreased storage costs by 12x by architecting cloud storage system with Google Cloud Firebase
- Designed mobile-first UX and developed adaptive mobile layout utilizing Figma and React for code editor
- Improved user workflow by engineering context sensitive autocomplete, linting, and syntax highlighting
- Future-proofed editor changes by implementing Vite-based test library, writing 130+ distinct unit tests

Founder | Conclusive Edge

Dec 2020 – Jun 2024

- Coached 300+ students across the nation in communication and logic for competitive speech & debate
- Hosted podcast about effective communication, interviewed guess speakers, 5 stars on Apple Podcasts

Software Research Assistant | Carnegie Mellon

Sep 2022 – Jun 2023

- Conducted NLP with R on how discussion of inflation in earnings calls affect abnormal stock returns
- Completed structured and unstructured topic models, fixed-effects OLS linear regression, cleaned data

Projects and Leadership

Software Engineer | Multi-node Parallel Monte Carlo Option Pricing in CUDA and MPI

Dec 2025

- Parallelized Adaptive Multilevel Monte Carlo with Milstein Discretization algorithm
- Achieved 178x speedup on NVIDIA RTX 2080 and 5,430x speedup on 4 NVIDIA Tesla V100s
- Maximized scalability via thread block memory, CUDA Thrust, and cuRAND optimization

Vice President | Tartan Neurotech

Dec 2022 – Jun 2023

- Led Python project utilizing electromyography muscle sensors to control a rover with Arduino
- Responsible for outreach to student body, big-picture vision, data analysis, and meeting leadership

Software Engineer | C Compiler

Dec 2023

- Architected compiler translating C into x86-64 bit assembly from scratch, written in Rust
- Designed lexing, parsing, and code generation pipeline to conduct lexical analysis and validation
- Implemented abstract syntax trees to support functions, loops, conditionals, variables, operators, more

Program Manager & Developer | Slash n Bash

Nov 2022

- Led team to win 1st place in Carnegie Mellon Hack112 out of 200 competitors with computer vision game
- Wrote game mechanics in Python, applied OpenCV, designed test cases, improved runtime by 17x

Languages and Technologies

Languages C++, SQL, Java, Rust, Python, C, Go, Typescript, Javascript, HTML/CSS

Technologies AWS, GCP, Azure, Kubernetes, Helm, Docker, Apache Spark, CUDA, OpenMP, MPI, React, Node.js, Firebase, MongoDB, Flask, Git, Sk-learn, Pandas, Numpy, Seaborn